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METHODS AND SYSTEMS FOR COLLECTING AND RELEASING VIRTUAL OBJECTS BETWEEN DISPARATE AUGMENTED REALITY ENVIRONMENTS

Abstract

Methods and systems are disclosed for collecting and releasing virtual objects between disparate augmented reality environments. One method comprises receiving user selection to collect an object in a first environment displayed on a user device at a first time. A search request may then be transmitted to a content server from which a virtual object corresponding to the collected object is received in response. The virtual object may be stored in a user library associated with a user identifier of the user. When a second environment is displayed on the user device, the virtual object may be added to the second environment in response to receiving user selection to add the virtual object and determining that the user is associated with the user identifier.

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS [0001] This application is a continuation of and claims the benefit of priority to U.S. application Ser. No. 18/330,782, filed on Jun. 7, 2023, which is a continuation of U.S. application Ser. No. 18/050,646, filed on Oct. 28, 2022, now U.S. Pat. No. 11,710,111, which is a continuation of U.S. application Ser. No. 16/685,171, Filed on Nov. 15, 2019, now U.S. Pat. No. 11,521,189, the contents of each of which are incorporated herein by reference in their entirety.

TECHNICAL FIELD

[0002] Various embodiments of the present disclosure relate generally to augmented reality user interfaces and, more particularly, to enabling electronic transactions and presenting virtual objects across disparate augmented reality environments.

BACKGROUND

[0003] Enhanced portability of consumer electronics with sophisticated graphics and computing capabilities (e.g., smart phones, tablets, and the like) have expanded the activities a user can engage in using those portable devices. The advancements in technologies have also brought science fiction into reality. For instance, a variety of sensing capabilities of portable devices can be utilized to further enhance the users' augmented reality experiences.

[0004] Augmented reality (AR) is the process of adding computer-supplied content, including images, video, text, and other data as layers over displayed images of the real world. For example, when a portable device captures an image or video of a scene including objects, one or more applications installed on the device can add information about the objects or other related contents over the displayed scene to enhance the user's experience. The user may utilize the overlaid information and contents to make informed decisions and/or to generate personal entertainment.

[0005] The background description provided herein is for the purpose of generally presenting the context of the disclosure. Unless otherwise indicated herein, the materials described in this section are not prior art to the claims in this application and are not admitted to be prior art, or suggestions of the prior art, by inclusion in this section.

SUMMARY OF THE DISCLOSURE

[0006] One embodiment provides a computer-implemented method for collecting and releasing virtual objects between disparate augmented reality (AR) environments, comprising: generating a user identifier of a user; displaying, at a first time, a first environment comprising an object; receiving user selection to collect the object; transmitting, to an AR content server, a search request for a virtual object corresponding to the object; receiving, from the AR content server, the virtual object; storing the virtual object in a user library associated with the user identifier; displaying, at a second time, a second environment; determining that the user is associated with the user identifier;

receiving user selection to add the virtual object to the second environment; and superimposing the virtual object over the second environment.

[0007] One embodiment provides a system for collecting and releasing virtual objects between disparate augmented reality (AR) environments. The system may comprise one or more processors; and a data storage comprising instructions which, when executed by the one or more processors, cause the one or more processors to perform a method comprising: generating a user identifier of a user; displaying, at a first time, a first environment comprising an object; receiving user selection to collect the object; transmitting, to an AR content server, a search request for a virtual object corresponding to the object; receiving, from the AR content server, the virtual object; storing the virtual object in a user library associated with the user identifier; displaying, at a second time, a second environment; determining that the user is associated with the user identifier; receiving user selection to add the virtual object to the second environment; and superimposing the virtual object over the second environment.

[0008] One embodiment provides a non-transitory computer readable medium for collecting and releasing virtual objects between disparate augmented reality (AR) environments. The non-transitory computer readable medium may store instructions which, when executed by one or more processors, cause the one or more processors to perform a method comprising: generating a user identifier of a user; displaying, at a first time, a first environment comprising an object; receiving user selection to collect the object; transmitting, to an AR content server, a search request for a virtual object corresponding to the object; receiving, from the AR content server, the virtual object; storing the virtual object in a user library associated with the user identifier; displaying, at a second time, a second environment; determining that the user is associated with the user identifier; receiving user selection to add the virtual object to the second environment; and superimposing the virtual object over the second environment.

[0009] Additional objects and advantages of the disclosed embodiments will be set forth in part in the description that follows, and in part will be apparent from the description, or may be learned by practice of the disclosed embodiments. The objects and advantages of the disclosed embodiments will be realized and attained by means of the elements and combinations particularly pointed out in the appended claims.

[0010] It is to be understood that both the foregoing general description and the following detailed description are exemplary and explanatory only and are not restrictive of the disclosed embodiments, as claimed.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0011] The accompanying drawings, which are incorporated in and constitute a part of this specification, illustrate various exemplary embodiments and together with the description, serve to explain the principles of the disclosed embodiments.

[0012] FIG. 1 depicts a block diagram of an augmented reality (AR)-enabled electronic transaction system, according to one aspect of the present disclosure.

[0013] FIG. 2A depicts an exemplary embodiment of a user device, according to one aspect of the present disclosure.

[0014] FIG. 2B depicts an exemplary embodiment of an AR content retrieval server, according to one aspect of the present disclosure.

[0015] FIG. 2C depicts an exemplary embodiment of a user library, according to one aspect of the present disclosure.

[0016] FIG. 3 illustrates an exemplary context in which the techniques presented in the present disclosure may be used.

[0017] FIGS. 4A-4J depict exemplary graphical user interfaces (GUIs) presented by a user device for AR-enabled transaction and user collection of virtual objects.

[0018] FIG. 5 is a flowchart of an exemplary method of collecting and releasing a virtual object between disparate environments, according to one aspect of the present disclosure.

[0019] FIG. 6 is a flowchart of an exemplary method of processing a search request received from an AR application at an AR content retrieval server, according to one aspect of the present disclosure.

[0020] FIG. 7 is a flowchart of an exemplary method of processing transaction information received from an AR application at a merchant system, according to one aspect of the present disclosure.

[0021] FIG. 8 is a flowchart of another exemplary method of processing a search request received from an AR application at an AR content retrieval server, according to one aspect of the present disclosure.

[0022] FIG. 9 illustrates an implementation of a computer system that may execute techniques presented herein.

DETAILED DESCRIPTION OF EMBODIMENTS

[0023] Various embodiments of the present disclosure relate generally to augmented reality user interface and, more particularly, to enabling electronic transactions and presenting virtual objects across disparate augmented reality environments.

[0024] The subject matter of the present disclosure will now be described more fully with reference to the accompanying drawings that show, by way of illustration, specific exemplary embodiments. An embodiment or implementation described herein as “exemplary” is not to be construed as preferred or advantageous, for example, over other embodiments or implementations; rather, it is intended to reflect or indicate that the embodiment(s) is/are “example” embodiment(s). Subject matter may be embodied in a variety of different forms and, therefore, covered or claimed subject matter is intended to be construed as not being limited to any exemplary embodiments set forth herein; exemplary embodiments are provided merely to be illustrative. Likewise, a reasonably broad scope for claimed or covered subject matter is intended. Among other things, for example, subject matter may be embodied as methods, devices, components, or systems. Accordingly, embodiments may, for example, take the form of hardware, software, firmware or any combination thereof (other than software per se). The following detailed description is, therefore, not intended to be taken in a limiting sense.

[0025] Throughout the specification and claims, terms may have nuanced meanings suggested or implied in context beyond an explicitly stated meaning. Likewise, the phrase “in one embodiment” as used herein does not necessarily refer to the same embodiment and the phrase “in another embodiment” as used herein does not necessarily refer to a different embodiment. It is intended, for example, that claimed subject matter include combinations of exemplary embodiments in whole or in part.

[0026] The terminology used below may be interpreted in its broadest reasonable manner, even though it is being used in conjunction with a detailed description of certain specific examples of the present disclosure. Indeed, certain terms may even be emphasized below; however, any terminology intended to be interpreted in any restricted manner will be overtly and specifically defined as such in this Detailed Description section.

[0027] For simplicity, the description that follows will be provided by reference to a “payment vehicle,” “transaction vehicle,” or a “payment card,” which generally refers to any type of financial alternative to cash. As is to be clear to those skilled in the art, no aspect of the present disclosure is specifically limited to a specific type of payment vehicle or payment card. Therefore, it is intended that the following description encompasses the use of the present disclosure with many other forms of financial alternatives to cash, including credit cards, debit cards, smart cards, chip-based payment cards, single-use cards, prepaid cards, electronic currency (such as might be provided

through a cellular telephone or personal digital assistant), and the like. Payment vehicles or payment cards can be traditional plastic transaction cards, titanium-containing, or other metal-containing, transaction cards, clear and/or translucent transaction cards, foldable or otherwise unconventionally-sized transaction cards, radio-frequency enabled transaction cards, or other types of transaction cards, such as credit, charge, debit, prepaid or stored-value cards, electronic benefit transfer cards, a “virtual” card (e.g., in the form of a display on a smart phone), a digital wallet, or any other like financial transaction instrument. In any event, the payment vehicles described herein communicate account information (e.g., an account number or other account indicative information) during a purchase event and/or payment or credit transaction.

[0028] The disclosed methods and systems relate to unique augmented reality (AR) features that may provide a more immersive user experience at public or private establishments, such as, for example, zoos, museums, theme parks, amusement parks, theatres, and so on. Adding a layer of additional reality may provide users with rich experiences and information, and may motivate users to stay engaged with the experience for an extended period of time. The disclosed methods and systems may also enable the user to save augmented reality content for future use and/or personal entertainment. In one embodiment, a user may be able to view additional information about an object detected in an environment, “collect” the object, and add a virtual version of the collected object (i.e., virtual object) to a user library. To collect the object, an electronic payment interface may be activated within the AR environment through which the user may make payments in a convenient manner. The user may be able “drop” the virtual object onto another environment to re-observe the object and/or for continued entertainment in a new setting.

[0029] Referring now to the appended drawings, FIG. 1 depicts a block diagram of an AR-enabled electronic transaction system **100**, according to one aspect of the present disclosure. In general, FIG. 1 depicts a payment environment **110**, a merchant system **150**, an AR content retrieval server **140**, and user device **130**, all connected via network **160**. Network **160** may include the Internet, but may also include other networks such as a corporate WAN, cellular network, satellite network, or combination thereof, for example. The network **160** may be employed to enable data communications between the various entities illustrated in FIG. 1.

[0030] In the AR-enabled electronic transaction system **100** of FIG. 1, a consumer (i.e., a user), during a checkout process with a merchant **150** (i.e., merchant system **150**), may make an electronic payment using a remote point of sale (POS) terminal **152**. The remote POS terminal **152** may comprise a server consistent with or similar to the computing system depicted in FIG. 9, and may handle transactions that occur over the network **160** with the user device **130**. The user device **130** may include, but may not be limited to, an AR consumer product (e.g., an AR headset, an AR eyewear, etc.), a smartphone, a mobile phone, a tablet computer, a laptop, a portable computer, a desktop computer, a smart watch, a wearable device, etc. The user device **130** may comprise a computing device consistent with or similar to the computing system depicted in FIG. 9. The remote POS terminal **152** may be in communication with the user device **130** over the network **160** and may provide an electronic payment interface to the user device **130**, using payment application **154**. The electronic payment interface may be provided in various formats. In one embodiment, the electronic payment interface may be a web page where a user of the user device **130** can provide payment vehicle information as well as the corresponding authentication data (e.g., PIN, password, CVV, answer to a secret question, biometrics data, etc.). In another embodiment, the electronic payment interface may be a web-based application through which a user can provide a payment vehicle (or payment vehicle information) as well as the corresponding authentication data. In yet another embodiment, the electronic payment interface may provide an installable mobile application to the user device **130**, so that a user may use the mobile application to access an electronic payment interface to provide a payment vehicle (or payment vehicle information) as well as the corresponding authentication data.

[0031] The AR-enabled electronic transaction system **100** may also comprise an AR content

retrieval server **140**. As will be discussed in greater detail below, the AR content retrieval server **140** may store various contents that could augment the scene or environment displayed by the user device **130**. For example, the user device **130** may capture and display an environment including one or more objects, and may request additional contents from the AR content retrieval server **140** to be presented over the displayed environment. The AR content retrieval server **140** may thus be configured to receive and process requests received from the user device **130**, retrieve contents from a storage, and transmit the contents to the user device **130**. To that end, the AR content retrieval server **140** may comprise a computing system consistent with, or similar to, that depicted in FIG. **9**. The AR content retrieval server **140** may provide additional functionalities, which will be discussed in greater detail in the following sections.

[0032] Because merchant **150** generally can use a bank or financial institution that is different from that of the consumer, an acquirer processor server(s) **115** may handle the electronic transactions between the financial institution of the consumer and that of the merchant **150**. Upon receiving a payment vehicle (or payment vehicle information) and the corresponding authentication data, the merchant system **150** (or the remote POS terminal **152** therein) may send a transaction authorization request by way of the network **160** to an acquirer processor server **115**. The acquirer processor server **115** may then request, by way of payment network **120**, an electronic transfer of funds to the financial institution **125** associated with the merchant **150**.

[0033] It should be noted that, although the merchant system **150**, AR content retrieval server **140**, and user device **130** are shown as separate entities in FIG. **1**, all or a portion of these components (and the steps performed thereof) may be implemented in a single computing device consistent with or similar to that depicted in FIG. **9**. For example, in one embodiment, the AR content retrieval server **140** and user device **130** may be implemented on a single computing device. In another embodiment, the AR content retrieval server **140** and merchant system **150** may be implemented on a single computing device. In yet another embodiment, the merchant system **150**, AR content retrieval server **140**, and user device **130** may indeed be implemented separately on their respective computing devices. The configurations specifically discussed herein regarding the arrangement of the components depicted in FIG. **1** are merely exemplary, and it would be obvious to a person of ordinary skill in the relevant art that different combinations of components may be implemented on a single or multiple computing devices.

[0034] FIG. **2A** depicts an exemplary embodiment of a user device **130** according to one aspect of the present disclosure. As disclosed above, the user device **130** may be a computing device consistent with or similar to that depicted in FIG. **9**. In addition, the user device **130** may comprise sensor(s) **232**, an AR application **234**, and a payment vehicle application **236**. The payment vehicle application **236** may provide security and encryption (e.g., tokenization, etc.) for personal information needed for electronic transactions. For example, the payment vehicle application **236** may comprise digital wallet systems such as GOOGLE PAY, APPLE PAY, SAMSUNG PAY, PAYPAL, VENMO, etc. In one embodiment and as shown in FIG. **2A**, the payment vehicle application **236** may be stored on the client side (i.e. user device **130**) and may be fully compatible with electronic commerce websites (i.e., merchant websites). In some embodiments, the payment vehicle application **236** may be on the server side, and may be one that a merchant or an organization creates for/about a user and maintains on its server(s). The personal information retained by the payment vehicle application **236** may comprise a shipping address, a billing address, payment methods, account numbers, expiration dates, security numbers and answers, and any other information needed for authenticating an electronic transaction.

[0035] The AR Application **234** may be a single application or a group of applications configured to collect various contextual data using one or more sensors **232** of the user device **130**, process the collected contextual data, and provide an AR experience to the user based on the processed contextual data. The sensor(s) **232** may include global positioning system (GPS) sensors, vision sensors (i.e., cameras), audio sensors (i.e., microphones), light sensors, temperature sensors, radio

frequency sensors, direction sensors (i.e., magnetic compasses, magnetometers, gyroscopes), and acceleration sensors (i.e., accelerometers). Using a camera of the user device **130**, the AR application **234** may receive a visual input that represents the real physical environment, and may identify objects in the captured environment. In one embodiment, the objects may be identified using a machine learning model. The machine learning model techniques used to detect the design elements may comprise any machine learning techniques suitable for detecting objects in visual contents/files, such as documents, images, pictures, drawings, media files, etc. However, other currently-known or later-developed object detection techniques may be used to identify the objects in the captured environment.

[0036] One of the machine learning techniques that may be useful and effective for object detection is a convolutional neural network (CNN), which is a type of supervised machine learning. In the CNN, an image data may be input and passed through various convolutional layers and pooling layers. The CNN applies a series of filters to the raw pixel data of the image to extract and learn higher-level features, which the model may then use for classification. Typically, a CNN may be composed of a stack of convolutional modules that perform feature extraction, with each module consisting of a convolutional layer followed by a pooling layer. The last convolutional module may be followed by one or more dense layers that perform classification. The types of the CNN which can be implemented with the embodiments contemplated herein may include a region-based CNN (RCNN), fast RCNN, and faster RCNN.

[0037] Nonetheless, it should be noted that other machine learning techniques and frameworks may be used to perform the object detection contemplated by the present disclosure. For example, the detection may be realized using other types of supervised machine learning such as regression problems, random forest, etc., using unsupervised machine learning such as cluster algorithms, principal component analysis (PCA), etc., and/or using reinforcement learning.

[0038] With renewed reference to FIG. 2A, the AR application **234** may receive additional contextual data from other sensors of the user device **130**, such as, e.g., the GPS sensor, accelerometer, and gyroscope, in order to determine the location, distance, angle, and/or motion of the identified objects relative to the camera of the user device **130**, or the location, distance, angle, and/or motion of the camera of the user device **130** relative to the identified objects. Data comprising the determined location, distance, angle, and/or motion may be referred to as location and orientation data in the present disclosure. The AR application **234** may then use the location and orientation data to adjust the size, orientation, and/or placement of virtual contents over the real-world environment displayed in the user device **130**. The AR application **234** may then overlay or superimpose the virtual contents over the displayed environment based on the adjusted parameters. The AR application **234** may request the virtual contents from a remote server (e.g., the AR content retrieval server **140**), and/or may retrieve the virtual contents from a local storage.

[0039] FIG. 2B depicts an exemplary embodiment of an AR content retrieval server **140** according to one aspect of the present disclosure. The AR content retrieval server **140** may be a computing device consistent with or similar to that depicted in FIG. 9. Additionally, the AR content retrieval server **140** may comprise an AR content manager **242** and an AR content storage **244**. The AR content manager **242** may be an application configured to receive a request for a virtual content from the AR application **234** of the user device **130**. Based on the request, the AR content manager **242** may search for the virtual content in the AR content storage **244**. Specifically, the AR content storage **244** may store object contents **246**, which may comprise virtual objects corresponding to the objects captured by the AR application **234** of the user device **130**, as well as detailed and/or related information pertaining to the captured objects. The AR content storage **244** may also store one or more user libraries **248** comprising virtual objects and/or icons corresponding to the virtual objects, each user library being associated with a respective user and comprising one or more virtual objects that belong to the user.

[0040] FIG. 2C depicts an exemplary embodiment of a user library **248** according to one aspect of

the present disclosure. As shown in FIG. 2C, each user library **248** may be a data structure storing a user identifier **250** in association with one or more virtual objects **252a**, **252b** that belong to the user. Each user library **248** may thus be associated with a user identifier **250** of the corresponding user. The AR content manager **242** may be configured to search for a library of virtual objects that belong to a particular user, using the user identifier of that user. Further, each virtual object (**252a** or **252b**) stored in the user library **248** may in turn be associated with a timestamp (**254a** or **254b**), a duration (**256a** or **256b**), and a corresponding icon (**258a** or **258b**). Use of these data (e.g., timestamp, duration, and icon) will be discussed in greater detail in the following sections.

[0041] FIG. 3 illustrates an exemplary context in which the techniques presented in the current disclosure may be used, according to one aspect of the present disclosure. For example, a user **305** may use the user device **130** to capture an image or real-time camera feed of an environment **300** including a lion **310** (i.e., an object **310**) at a zoo. The AR application **234** of the user device **130** may augment the displayed environment with additional content pertaining to the lion **310**. The types of the additional content and the manner in which the real-world environment is augmented are explained in detail in the following sections. For example, the AR application **234** may be used in conjunction with the payment vehicle application **236** and merchant system **150** to provide an AR-enabled transaction interface. Furthermore, the AR application **234** may be used in conjunction with the AR content retrieval server **140** to enable users to collect virtual objects corresponding to real-world objects detected in the environment.

[0042] FIGS. 4A-4J depict exemplary graphical user interfaces (GUIs) presented by the user device **130** for AR-enabled transaction and user collection of virtual objects. Specifically, FIGS. 4A-4J illustrate the GUIs that the AR application **234** working in conjunction with one or more of the payment vehicle application **236**, merchant system **150**, and AR content retrieval server **140** may render on the user device **130**. Further, FIGS. 4A-4F illustrate GUIs rendered by the AR application **234** at a first environment, and FIGS. 4G-4J illustrate GUIs rendered by the AR application **234** at a second environment.

[0043] In order to use the AR features contemplated in the current disclosure, the user **305** may need to start and/or login to the AR application **234** to start capturing the surrounding environment for an AR experience. The user **305** may provide login credentials such as, for example, login ID and password, user biometrics data (e.g., fingerprint, iris patterns, facial geometry, voice, etc.), etc. and the login credentials may be used to generate a user identifier associated with that user. In some embodiments, the login credentials themselves may be used as the user identifier. In other embodiments, the login procedure may not be necessary. Instead, the AR application **234** may generate the user identifier based on user biometrics data that are automatically captured using one or more sensors **232** of the user device **130**. This way, the user's identity may be verified without requiring the login procedure, by monitoring the user's biometrics data using one or more sensors **232** of the user device **130** periodically, continuously, and/or upon detecting a triggering event (e.g., user's face/eye faces the camera, user's finger touches the fingerprint scanner, etc.). The user biometrics data captured automatically in lieu of the login procedure may be referred to as automatically-captured user biometrics data throughout the present disclosure. Furthermore, if the user **305** is not a first time user, instead of generating a new user identifier, the AR application **234** may identify a previously-generated user identifier of the user **305** based on the login credentials or automatically-captured user biometrics data.

[0044] In FIG. 4A, the AR application **234** may receive a visual input from a camera of the user device **130**, the visual input representing a first environment **300**. The AR application **234** may employ object detection techniques to detect the lion **310** (i.e., detect the object **310** and the object type "lion") within the displayed environment **300**. Once the AR application **234** detects the lion **310**, the AR application **234** may transmit a search request containing the object type (e.g., lion) to the AR content retrieval server **140**. The AR content retrieval server **140** may retrieve additional content pertaining to the object type "lion," and transmit the additional content back to the AR

application **234**. For example, the additional content may comprise information related to items that the user **305** can purchase in order to “sponsor” the lion, and any other information relevant to the lion.

[0045] Concurrently to the search step or in response to receiving the additional content from the AR content retrieval server **140**, the AR application **234** may determine a position within the display at which the additional content pertaining to the lion **310** may be superimposed/overlaid. As described above, the position may be determined based on contextual data received from one or more sensors of the user device **130**, such as a GPS sensor, an accelerometer, a gyroscope, etc. If the additional content comprises discrete portions of information which should be superimposed over the displayed environment **300** separately, the position for each portion of information may be determined based on the contextual data and position determination criterion specific to that portion of information. The aforementioned position determination algorithm may also be used to determine positions of any other user interface elements that may be superimposed/overlaid, such as options or instructions provided to the user **305**.

[0046] Upon receiving the additional content from the AR content retrieval server **140** (or even prior to receiving the additional content), the AR application **234** may superimpose an option **405** to sponsor the lion **310** at a determined position. The position of the option **405** may be adjusted in real-time based on the contextual data being received iteratively as the user device **130** and/or the detected object **310** moves. In other words, by continuously receiving, processing, and analyzing the visual input and contextual data, the AR application **234** can keep the superimposed/overlaid content adequately aligned with the lion **310**. Such real-time adjustment may be applicable to any type of content that is superimposed over a real-world environment (i.e., not limited to the option **405**). The user **305** may elect to sponsor the lion **310** by touching or clicking on the option **405**, or via voice recognition. For example, the user device **130** may be equipped with a voice recognition application capable of determining whether the user **305** selects or declines the option **405**. For example, the user **305** dictating “Sponsor the lion” may mean that the user **305** has selected the option **405**.

[0047] In FIG. **4B**, upon the user selecting the option **405**, the AR application **234** may display sponsor items that the user **305** can purchase. More specifically, the sponsor items such as, for example, meat **410b** and tree **415b**, may be superimposed at appropriate positions determined via the position determination algorithm. Further, to receive user selection to purchase the sponsor items **410b** and **415b**, the AR application **234** may also superimpose “buy” options **410a** and **415a** at appropriate positions determined via the position determination algorithm.

[0048] In FIG. **4C**, upon the user selecting the option **410a** to purchase the meat **410b**, the AR application **234** may provide an electronic payment interface via the merchant system **150**. As described above, the merchant system **150** may comprise a remote POS terminal **152**, which is in communication with the user device **130** and is configured to provide an electronic payment interface to the user device **130** using payment application **154**. Although the electronic payment interface may be any one of a web page, web-based application, or an installable mobile application as discussed above, the payment vehicle application **236** of the user device **130** may work in conjunction with the remote POS terminal **152** to carry out an electronic transaction. For example, upon the user selecting the “buy” option **410a**, rather than providing an electronic payment interface by way of a web page, web-based application, or an installable mobile application, the payment vehicle application **236** compatible with the remote POS terminal **152** may be initiated. As shown in FIG. **4C**, by utilizing data provided from the payment vehicle application **236** (e.g., a digital wallet application), the AR application **234** may superimpose an electronic payment window **410c** through which the user **305** may confirm/modify the displayed transaction information (e.g., user information comprising user name, user id, payment account number, etc., which may be retained by the payment vehicle application **236** securely, transaction amount, etc.) and complete the payment in a more convenient and prompt manner. Upon the user

selecting the “Pay now” button, the AR application **234** may, by way of the payment vehicle application **236**, transmit the transaction information to the merchant system **150** (or the remote POS terminal **152** of the merchant system **150**). The merchant system **150** may then generate a transaction authorization request based on the received transaction information, and transmit the transaction authorization request to an acquirer processor server **115**.

[0049] User may continue to purchase sponsor items as shown in FIG. **4D**. FIG. **4D** shows another electronic payment window **415c** superimposed over the environment **300**, upon the user selecting the option **415a**. As discussed above, the electronic payment window **415c** may be generated and superimposed by the AR application **234**, based on data provided from the payment vehicle application **236**.

[0050] Once an amount equal to or exceeding a predetermined threshold amount has been spent by the user **305** on sponsor items, the AR application **234** may provide the user **305** with an option **420** to “collect” the lion **310**, as shown in FIG. **4E**. In some embodiments, the AR application **234** may provide the option **420** to “collect” the lion **310** regardless of the amount spent on sponsor items or whether the user has purchased any sponsor items. However, there may be other restrictions or conditions applied against the user **305** if the user **305** is allowed to collect the lion **310** regardless, which will be described in detail in later sections.

[0051] If the user elects to “collect” the lion **310** by touching or clicking on the option **420**, the AR application **234** may transmit a search request to the AR content retrieval server **140**, the search request including the object type previously determined using object detection techniques, along with a user identifier associated with the user **305**. This time however, the search request may be to retrieve a virtual object corresponding to the real-world object **310** (e.g., a lion). Such a virtual object may be a three-dimensional or two-dimensional virtual character representative of the real-world object **310**. In some embodiments, the virtual object may merely be a three-dimensional rendering of the detected real-world object **310**, which may be identical or highly similar to the detected object **310**. In other embodiments, the virtual object may be a three-dimensional or two-dimensional animated character representative of the detected object **310**.

[0052] As described above, virtual objects corresponding to various real-world objects may be stored in the AR content storage **244** of the AR content retrieval server **140**. Upon receiving the search request for the virtual object, the AR content manager **242** may use the object type indicated in the search request to retrieve the virtual object associated with the object type and/or an icon corresponding to the virtual object from the AR content storage **244**. The AR content manager **242** may then use the user identifier contained in the search request to determine whether a user library associated with the user **305** exists in the AR content storage **244**. If the user library for the user **305** already exists in the AR content storage **244**, the AR content manager **242** may add the retrieved virtual object and/or icon to the user library, if the retrieved virtual object and/or icon do not already exist in the user library. A user library may thus include more than one virtual object and/or icon. If the user library for the user **305** does not exist in the AR content storage **244**, the AR content manager **242** may generate a user library, associate the user library with the user identifier contained in the search request, and add the retrieved virtual object and/or icon to the user library. The newly-generated user library may then be stored in the AR content storage **244**. In essence, a user library links virtual object(s) retrieved for a user (i.e., virtual object(s) that belong to a user) and corresponding icon(s) with a user identifier of that user.

[0053] The entire user library, or one or more virtual objects and/or icons contained in the user library, may then be returned to the AR application **234** of the user device **130**. The virtual object(s) and corresponding icon(s) may then be stored in association with the user identifier in a local storage of the user device **130**. The virtual object(s) and corresponding icon(s) stored at the user device **130** may thus resemble the user library stored in the AR content storage **244**. In other words, the data stored in the local storage of the user device **130** may be in sync with the user library stored in the AR content storage **244**, such that it may be unnecessary for the AR application **234** to

contact the AR content storage **244** every time a virtual object and/or icon are to be rendered/displayed by the AR application **234**.

[0054] In some embodiments, the virtual objects and corresponding icons associated with the user **305** (i.e., stored in the user library) may be valid only for a certain amount of time depending on the amount of money spent by the user **305** on sponsor items. For instance, each virtual object and its corresponding icon may be associated with a timestamp indicative of the date they were created and/or added to the user library, and another timestamp or value that indicates how long the virtual object and the icon are valid for. In another embodiment, each virtual object and its corresponding icon may be associated with an expiration date and/or time. The timestamp(s) and/or expiration date/time may be stored in the user library in association with the corresponding virtual character and/or icon. To that end, the search request for the virtual object may contain, in addition to the user identifier and object type, a transaction amount, which the AR content manager **242** may use to determine the timestamp(s) and/or expiration date/time for the virtual object. Based on the timestamp(s) and/or expiration date/time, the AR application **234** may monitor when each virtual object expires, and provides a notification to the user when the virtual object expires. The AR application **234** may also display an option for the user **305** to purchase additional sponsor items (in manners similar to those illustrated in FIGS. **4B-4D**) and modify the timestamp(s) and/or expiration date/time based on the additional purchase(s). If the user decides not to purchase additional sponsor items, the virtual object may become expired and be removed from the user library.

[0055] The AR application **234** may display a side pane or side bar **425** containing the icon(s) **430a-430e** of the virtual object(s) that have been collected by the user **305**. From a user experience standpoint, upon the user **305** selecting the option **420** to collect the lion **310**, the side pane **425** may appear briefly to confirm collection by showing the icon **430a** (along with icon(s) representative of any other virtual object(s) that have been collected by the user **305**) and may fold back to the side of the screen in order not to obstruct the entire screen view. However, a small tab may remain on the side of the screen, which the user **305** can touch and swipe to the other side to re-open/expand the side pane **425**.

[0056] In FIG. **4G**, the AR application **234** may receive a visual input from the camera of the user device **130**, the visual input representing a second environment **400**. The second environment **400** may be the same or different from the first environment **300**. In FIG. **4G**, the second environment **400** is shown to be different from the first environment **300**. For example, the second environment **400** may be an interior of a home while the first environment was an enclosure at a zoo. At the second environment **400**, the user **305** may have logged out of the AR application **234** and logged back in, which may have required the user **305** to provide the login credentials again. In this case, the AR application **234** may receive the login credentials, and verify that the user **305** is associated with a user identifier based on the login credentials. The AR application **234** may then retrieve the user identifier of the user **305**. Using the user identifier, the AR application **234** may retrieve the user library containing the virtual object(s) collected by the user **305** and corresponding icon(s) from the local storage (or from the AR content storage **244**, alternatively), and display the side pane **425** containing the icon(s) **430a-430e**. In case the user remains logged in from the first to the second environment, the steps for verifying the user identify based on login credentials may not be necessary. Further, in case the AR application **234** does not require the login procedure and may instead rely on automatically-captured user biometrics data for user verification, the AR application **234** may receive user biometrics data from one or more sensor(s) of the user device **130**, use the received user biometrics data to find a matching user identifier. Using the user identifier, the AR application **234** may retrieve the user library and display the side pane **425** containing the icon(s) **430a-430e**. FIG. **4H** shows the side pane **425** displayed over the second environment **400**.

[0057] As shown in FIG. **4I**, the user **305** may then select an icon, for instance the icon **430a** from the side pane **425**, and drag and drop the icon **430a** onto the second environment **400**. Once the

icon **430a** is dropped onto the second environment **400**, the icon **430a** may turn into a virtual object **440** with or without transition animations, as shown in FIG. **4J**. As described above, the virtual object **440** may be a two-dimensional or three-dimensional rendering or animated character of the real-world object **310**. The virtual object **440** added to the second environment **400** may be mobile within the captured environment **400** and may perform certain movements or expressions to the entertainment of the user **305**. Further, the user **305** may be able to select the virtual object **440** to view additional information such as, for example, sponsor items and amount that have been donated, general information about the virtual object **440** (i.e., object **310**), the location at which the virtual object **440** (i.e., object **310**) was collected, remaining usage time, related objects that the user **305** may be interested, etc.

[0058] In some embodiments, such as when the user **305** was previously given the option to collect the lion **310** without making any purchase, the user **305** may be presented with an option to purchase one or more sponsor items (in manners similar to those illustrated in FIGS. **4B-4D**) when the user **305** attempts to drop the icon **430a-430f** onto the second environment **400**. If the user **305** decides not to make any purchase at that point, the AR application **234** may present a notification to the user **305** that purchase of a sponsor item is needed to add the virtual object to the second environment **400**. If the user **305** makes a purchase, the AR application **234** may then add the virtual object to the second environment **400**.

[0059] FIG. **5** is a flowchart of an exemplary method of collecting and releasing a virtual object between disparate environments, according to one aspect of the present disclosure. Notably, method **500** may be performed by the AR application **234** of the user device **130**. In some embodiments, method **500** may be performed by the AR application **234** of the user device **130**, in part in conjunction with the payment vehicle application **236**. At step **510**, the AR application **234** may receive login credential(s) or automatically-captured user biometrics data of the user. At step **512**, the AR application **234** may generate a user identifier of the user based on the login credential(s) or automatically-captured user biometrics data. If the user is not a first-time user however, the AR application **234** may retrieve a previously-generated user identifier based on the login credential(s) or automatically-captured user biometrics data. At step **514**, the AR application **234** may display a first environment (e.g., first environment **300** illustrated in FIGS. **4A-4F**) based on a visual input received from a camera of the user device **130**. At step **516**, the AR application **234** may identify an object in the first environment using object detection techniques such as, for example, convolutional neural networks. At step **518**, the AR application **234** may determine the type of the identified object (i.e., object type). The type may be identified by a trained machine learning model such as the convolutional neural networks discussed above, which may be configured to distinguish between various objects based on visual features and classify each object under a object type or object name. At step **520**, the AR application **234** may transmit, to a server (i.e., AR content retrieval server **140** illustrated in FIG. **2B**), a search request for content related to the object, the search request including the object type.

[0060] The search request may be further processed by the AR content retrieval server **140** before a search response is returned to the AR application **234**. FIG. **6** is a flowchart of an exemplary method of processing the search request received from the AR application **234**, according to one aspect of the present disclosure. At step **610**, the AR content manager **242** may receive, from the user device **130**, the search request for content pertaining to the object. As discussed above, the search request may indicate the type of the object. At step **615**, the AR content manager **242** may search the AR content storage **244** for content pertaining to the object, using the object type indicated in the search request. Specifically, the content may be AR content that may be overlaid or superimposed over a displayed environment. At step **620**, the AR content manager **242** may retrieve the content pertaining to the object from the AR content storage **244**. At step **625**, the AR content manager **242** may transmit the retrieved content back to the user device **130**.

[0061] With renewed reference to FIG. **5**, at step **522**, the AR application **234** may receive the

content pertaining to the object from the AR content retrieval server **140**. At step **524**, the AR application **234** may overlay/superimpose at least a portion of the received content over the first environment. As discussed above in reference to FIGS. **4A-4B**, the overlaid/superimposed content may include additional information about the identified object, an option to sponsor the object, sponsor item(s), option(s) to purchase sponsor item(s), etc. At step **526**, the AR application **234** may receive user selection to initiate an electronic transaction. For example, as discussed above in reference to FIGS. **4B-4C**, the user may select the option to purchase a sponsor item. At step **528**, the AR application **234** may present a transaction interface (e.g., electronic payment window **410c**, **415c** in FIGS. **4C-4D**). As described above, the AR application **234** may work in conjunction with the payment vehicle application **236** to present the transaction interface. The transaction interface may allow the user to confirm/modify the displayed transaction information and make an electronic payment. At step **530**, the AR application **234** may receive transaction information via the transaction interface. At step **535**, the AR application **234** may transmit the transaction information to a merchant system (i.e., merchant system **150** in FIG. **1**).

[0062] The transaction information may be received and processed by the merchant system **150** to obtain a transaction authorization from a financial institution. FIG. **7** is a flowchart of an exemplary method of processing the transaction information at the merchant system **150**, according to one aspect of the present disclosure. At step **710**, the merchant system **150** may receive the transaction information from the user device **130**. At step **715**, the merchant system **150** may transmit a transaction authorization request to an acquirer processor server (i.e., acquirer processor server **115** in FIG. **1**). The transaction authorization request may be generated based on the transaction information received from the user device **130**, and may comprise information needed by downstream entities (e.g., acquirer processor server **115**, payment network **120**, financial institution **125**) to route and/or authorize the transaction. At step **720**, the merchant system **150** may receive a transaction authorization response from the acquirer processor server **115**. At step **725**, the merchant system **150** may transmit the transaction authorization response to the user device **130**.

[0063] With renewed reference to FIG. **5**, at step **534**, the AR application **234** may receive the transaction authorization response from the merchant system **150**. At step **536**, the AR application **234** may overlay/superimpose an option to collect the object over the first environment (e.g., option **420** in FIG. **4E**). At step **538**, the AR application **234** may receive user selection to collect the object. At step **540**, the AR application **234** may transmit a search request for a virtual object to the AR content retrieval server **140**. The search request may include the user identifier and object type determined in previous steps **512** and **518**, respectively. In some embodiments, the search request may additionally include a transaction amount based on the sponsor item(s) purchased.

[0064] The search request may be further processed by the AR content retrieval server **140** before a search response is returned to the AR application **234**. FIG. **8** is a flowchart of an exemplary method of processing the search request received from the AR application **234**, according to one aspect of the present disclosure. At step **810**, the AR content manager **242** may receive the search request from the user device **130**. As discussed above, the search request may include the user identifier of the user and the object type. At step **815**, the AR content manager **242** may search the AR content storage **244** for a virtual object corresponding to the object. The AR content manager **242** may use the object type indicated in the search request to retrieve the virtual object from the AR content storage **244**. At step **820**, the AR content manager **242** may retrieve the virtual object from the AR content storage **244**. The AR content manager **242** may also retrieve an icon corresponding to the virtual object from the AR content storage **244**. At step **825**, the AR content manager **242** may add the virtual and/or corresponding icon to a user library associated with the user.

[0065] As discussed above, the AR content manager **242** may first determine whether a user library associated with the user identifier already exists in the AR content storage **244**. Based on this determination, the user library may be retrieved from the AR content storage **244** if there is a

previously-generated user library, or the user library may be newly generated if there is no previously-generated user library associated with the user. The user library may associate the user identifier of the user with the virtual object and/or corresponding icon. In some embodiments, the AR content manager **242** may also determine timestamp(s) or expiration date/time indicative of how long the virtual object is valid for, based on the transaction amount included in the search request. Then, the AR content manager **242** may store the timestamp(s) or expiration date/time in association with the corresponding virtual object within the user library.

[0066] At step **830**, the AR content manager **242** may transmit the retrieved virtual object to the user device **130**. Alternatively, the AR content manager **242** may send the entire user library associated with the user to the user device **130**, if the user device **130** does not already have a copy of the user library stored in a local storage. In other words, if the user device **130** already has a local copy of the user library, the AR content manager **242** may only transmit the retrieved virtual object and/or corresponding icon to the user device **130**, such that the user device **130** may update the local copy with the retrieved virtual object and/or corresponding icon. On the other hand, if the user device **130** does not have a local copy of the user library, the AR content manager **242** may transmit the entire user library to the user device **130**, such that the user device **130** may retain a copy of the user library locally, which may be updated incrementally as additional virtual objects are collected by the user.

[0067] With renewed reference to FIG. 5, at step **542**, the AR application **234** may receive the virtual object, or the user library containing the virtual object if the user device **130** does not already have a local copy of the user library, from the AR content retrieval server **140**. At step **544**, the AR application **234** may store the virtual object in a user library associated with the user identifier in the local storage. As alluded to above, if the local storage does not have a copy of the user library, the AR application **234** may instead receive the entire user library containing the virtual object from the AR content retrieval server **140** and store the received user library as a local copy.

[0068] At step **546**, the AR application **234** may display a second environment (e.g., second environment **400** illustrated in FIGS. 4G-4J) based on a visual input received from the camera of the user device **130**. At step **548**, the AR application **234** may determine that the current user is associated with the user identifier generated at step **512** (meaning the user has remained the same). As discussed above in reference to FIGS. 4G-4H, the determination may be based on matching login credentials or automatically-captured user biometrics data (or a user identifier derived therefrom) of the current user to the user identifier generated at step **512**.

[0069] It should be noted that method **500** may still proceed to subsequent steps even if the AR application **234** determines that the current user is not associated with the user identifier generated at step **512** (meaning the user has changed). If the AR application **234** determines that the current user's login credentials or automatically-captured biometrics data (or a user identifier derived therefrom) do not match with the user identifier generated at step **512**, the AR application **234** may determine a user identifier associated with the current user based on the current user's login credentials or automatically-captured user biometrics data. That user identifier may be used to locate the corresponding user library from which virtual object(s) collected by the current user may be retrieved.

[0070] Once the AR application **234** determines that the user is associated with the user identifier, at step **550**, the AR application **234** may retrieve the user library associated with the user identifier. At step **552**, the AR application may present, over the second environment, a menu comprising one or more icons corresponding to one or more virtual objects stored in the user library. The one or more icons may comprise an icon corresponding to the virtual object collected by the user at step **538**. At step **554**, the AR application **234** may receive user selection to add the virtual object to the second environment. The user selection may comprise the user dragging the icon corresponding the virtual object to a desired location within the second environment, and dropping the icon onto the

desired location, as discussed above in reference to FIG. 4I. Responsive to this user selection, at step 556, the AR application 234 may add the virtual object to the second environment. In other words, the virtual object may be overlaid or superimposed over the displayed second environment, as discussed above in reference to FIG. 4J.

[0071] Unless specifically stated otherwise, as apparent from the following discussions, it is appreciated that throughout the specification, discussions utilizing terms such as “processing,” “computing,” “calculating,” “determining”, analyzing” or the like, refer to the action and/or processes of a computer or computing system, or similar electronic computing device, that manipulate and/or transform data represented as physical, such as electronic, quantities into other data similarly represented as physical quantities.

[0072] In a similar manner, the term “processor” may refer to any device or portion of a device that processes electronic data, e.g., from registers and/or memory to transform that electronic data into other electronic data that, e.g., may be stored in registers and/or memory. A “computer,” a “computing machine,” a “computing platform,” a “computing device,” or a “server” may include one or more processors.

[0073] FIG. 9 illustrates an implementation of a general computer system designated 900. The computer system 900 can include a set of instructions that can be executed to cause the computer system 900 to perform any one or more of the methods or computer based functions disclosed herein. The computer system 900 may operate as a standalone device or may be connected, e.g., using a network, to other computer systems or peripheral devices.

[0074] In a networked deployment, the computer system 900 may operate in the capacity of a server or as a client user computer in a server-client user network environment, or as a peer computer system in a peer-to-peer (or distributed) network environment. The computer system 900 can also be implemented as or incorporated into various devices, such as a personal computer (PC), a tablet PC, a set-top box (STB), a personal digital assistant (PDA), a mobile device, a palmtop computer, a laptop computer, a desktop computer, a communications device, a wireless telephone, a land-line telephone, a control system, a camera, a scanner, a facsimile machine, a printer, a pager, a personal trusted device, a web appliance, a network router, switch or bridge, or any other machine capable of executing a set of instructions (sequential or otherwise) that specify actions to be taken by that machine. In a particular implementation, the computer system 900 can be implemented using electronic devices that provide voice, video, or data communication. Further, while a single computer system 900 is illustrated, the term “system” shall also be taken to include any collection of systems or sub-systems that individually or jointly execute a set, or multiple sets, of instructions to perform one or more computer functions.

[0075] As illustrated in FIG. 9, the computer system 900 may include a processor 902, e.g., a central processing unit (CPU), a graphics processing unit (GPU), or both. The processor 902 may be a component in a variety of systems. For example, the processor 902 may be part of a standard personal computer or a workstation. The processor 902 may be one or more general processors, digital signal processors, application specific integrated circuits, field programmable gate arrays, servers, networks, digital circuits, analog circuits, combinations thereof, or other now known or later developed devices for analyzing and processing data. The processor 902 may implement a software program, such as code generated manually (i.e., programmed).

[0076] The computer system 900 may include a memory 904 that can communicate via a bus 908. The memory 904 may be a main memory, a static memory, or a dynamic memory. The memory 904 may include, but is not limited to computer readable storage media such as various types of volatile and non-volatile storage media, including but not limited to random access memory, read-only memory, programmable read-only memory, electrically programmable read-only memory, electrically erasable read-only memory, flash memory, magnetic tape or disk, optical media and the like. In one implementation, the memory 904 includes a cache or random-access memory for the processor 902. In alternative implementations, the memory 904 is separate from the processor 902,

such as a cache memory of a processor, the system memory, or other memory. The memory **904** may be an external storage device or database for storing data. Examples include a hard drive, compact disc (“CD”), digital video disc (“DVD”), memory card, memory stick, floppy disc, universal serial bus (“USB”) memory device, or any other device operative to store data. The memory **904** is operable to store instructions executable by the processor **902**. The functions, acts or tasks illustrated in the figures or described herein may be performed by the programmed processor **902** executing the instructions stored in the memory **904**. The functions, acts or tasks are independent of the particular type of instructions set, storage media, processor or processing strategy and may be performed by software, hardware, integrated circuits, firm-ware, micro-code and the like, operating alone or in combination. Likewise, processing strategies may include multiprocessing, multitasking, parallel processing and the like.

[0077] As shown, the computer system **900** may further include a display unit **910**, such as a liquid crystal display (LCD), an organic light emitting diode (OLED), a flat panel display, a solid-state display, a cathode ray tube (CRT), a projector, a printer or other now known or later developed display device for outputting determined information. The display **910** may act as an interface for the user to see the functioning of the processor **902**, or specifically as an interface with the software stored in the memory **904** or in the drive unit **906**.

[0078] Additionally or alternatively, the computer system **900** may include an input device **912** configured to allow a user to interact with any of the components of system **900**. The input device **912** may be a number pad, a keyboard, or a cursor control device, such as a mouse, or a joystick, touch screen display, remote control, or any other device operative to interact with the computer system **900**.

[0079] The computer system **900** may also or alternatively include a disk or optical drive unit **906**. The disk drive unit **906** may include a computer-readable medium **922** in which one or more sets of instructions **924**, e.g. software, can be embedded. Further, the instructions **924** may embody one or more of the methods or logic as described herein. The instructions **924** may reside completely or partially within the memory **904** and/or within the processor **902** during execution by the computer system **900**. The memory **904** and the processor **902** also may include computer-readable media as discussed above.

[0080] In some systems, a computer-readable medium **922** includes instructions **924** or receives and executes instructions **924** responsive to a propagated signal so that a device connected to a network **160** can communicate voice, video, audio, images, or any other data over the network **160**. Further, the instructions **924** may be transmitted or received over the network **160** via a communication port or interface **920**, and/or using a bus **908**. The communication port or interface **920** may be a part of the processor **902** or may be a separate component. The communication port **920** may be created in software or may be a physical connection in hardware. The communication port **920** may be configured to connect with a network **160**, external media, the display **910**, or any other components in system **900**, or combinations thereof. The connection with the network **160** may be a physical connection, such as a wired Ethernet connection or may be established wirelessly as discussed below. Likewise, the additional connections with other components of the system **900** may be physical connections or may be established wirelessly. The network **160** may alternatively be directly connected to the bus **908**.

[0081] While the computer-readable medium **922** is shown to be a single medium, the term “computer-readable medium” may include a single medium or multiple media, such as a centralized or distributed database, and/or associated caches and servers that store one or more sets of instructions. The term “computer-readable medium” may also include any medium that is capable of storing, encoding, or carrying a set of instructions for execution by a processor or that cause a computer system to perform any one or more of the methods or operations disclosed herein. The computer-readable medium **922** may be non-transitory, and may be tangible.

[0082] The computer-readable medium **922** can include a solid-state memory such as a memory

card or other package that houses one or more non-volatile read-only memories. The computer-readable medium **922** can be a random-access memory or other volatile re-writable memory. Additionally or alternatively, the computer-readable medium **922** can include a magneto-optical or optical medium, such as a disk or tapes or other storage device to capture carrier wave signals such as a signal communicated over a transmission medium. A digital file attachment to an e-mail or other self-contained information archive or set of archives may be considered a distribution medium that is a tangible storage medium. Accordingly, the disclosure is considered to include any one or more of a computer-readable medium or a distribution medium and other equivalents and successor media, in which data or instructions may be stored.

[0083] In an alternative implementation, dedicated hardware implementations, such as application specific integrated circuits, programmable logic arrays and other hardware devices, can be constructed to implement one or more of the methods described herein. Applications that may include the apparatus and systems of various implementations can broadly include a variety of electronic and computer systems. One or more implementations described herein may implement functions using two or more specific interconnected hardware modules or devices with related control and data signals that can be communicated between and through the modules, or as portions of an application-specific integrated circuit. Accordingly, the present system encompasses software, firmware, and hardware implementations.

[0084] The computer system **900** may be connected to one or more networks **160**. The network **160** may define one or more networks including wired or wireless networks. The wireless network may be a cellular telephone network, an 802.11, 802.16, 802.20, or WiMAX network. Further, such networks may include a public network, such as the Internet, a private network, such as an intranet, or combinations thereof, and may utilize a variety of networking protocols now available or later developed including, but not limited to TCP/IP based networking protocols. The network **160** may include wide area networks (WAN), such as the Internet, local area networks (LAN), campus area networks, metropolitan area networks, a direct connection such as through a Universal Serial Bus (USB) port, or any other networks that may allow for data communication. The network **160** may be configured to couple one computing device to another computing device to enable communication of data between the devices. The network **160** may generally be enabled to employ any form of machine-readable media for communicating information from one device to another. The network **160** may include communication methods by which information may travel between computing devices. The network **160** may be divided into sub-networks. The sub-networks may allow access to all of the other components connected thereto or the sub-networks may restrict access between the components. The network **160** may be regarded as a public or private network connection and may include, for example, a virtual private network or an encryption or other security mechanism employed over the public Internet, or the like.

[0085] In accordance with various implementations of the present disclosure, the methods described herein may be implemented by software programs executable by a computer system. Further, in an exemplary, non-limited implementation, implementations can include distributed processing, component/object distributed processing, and parallel processing. Alternatively, virtual computer system processing can be constructed to implement one or more of the methods or functionality as described herein.

[0086] Although the present specification describes components and functions that may be implemented in particular implementations with reference to particular standards and protocols, the disclosure is not limited to such standards and protocols. For example, standards for Internet and other packet switched network transmission (e.g., TCP/IP, UDP/IP, HTML, HTTP) represent examples of the state of the art. Such standards are periodically superseded by faster or more efficient equivalents having essentially the same functions. Accordingly, replacement standards and protocols having the same or similar functions as those disclosed herein are considered equivalents thereof.

[0087] It will be understood that the steps of methods discussed are performed in one embodiment by an appropriate processor (or processors) of a processing (i.e., computer) system executing instructions (computer-readable code) stored in storage. It will also be understood that the invention is not limited to any particular implementation or programming technique and that the invention may be implemented using any appropriate techniques for implementing the functionality described herein. The invention is not limited to any particular programming language or operating system.

[0088] It should be appreciated that in the above description of exemplary embodiments of the invention, various features of the invention are sometimes grouped together in a single embodiment, figure, or description thereof for the purpose of streamlining the disclosure and aiding in the understanding of one or more of the various inventive aspects. This method of disclosure, however, is not to be interpreted as reflecting an intention that the claimed invention requires more features than are expressly recited in each claim. Rather, as the following claims reflect, inventive aspects lie in less than all features of a single foregoing disclosed embodiment. Thus, the claims following the Detailed Description are hereby expressly incorporated into this Detailed Description, with each claim standing on its own as a separate embodiment of this invention.

[0089] Furthermore, while some embodiments described herein include some but not other features included in other embodiments, combinations of features of different embodiments are meant to be within the scope of the invention, and form different embodiments, as would be understood by those skilled in the art. For example, in the following claims, any of the claimed embodiments can be used in any combination.

[0090] Furthermore, some of the embodiments are described herein as a method or combination of elements of a method that can be implemented by a processor of a computer system or by other means of carrying out the function. Thus, a processor with the necessary instructions for carrying out such a method or element of a method forms a means for carrying out the method or element of a method. Furthermore, an element described herein of an apparatus embodiment is an example of a means for carrying out the function performed by the element for the purpose of carrying out the invention.

[0091] In the description provided herein, numerous specific details are set forth. However, it is understood that embodiments of the invention may be practiced without these specific details. In other instances, well-known methods, structures and techniques have not been shown in detail in order not to obscure an understanding of this description.

[0092] Similarly, it is to be noticed that the term coupled, when used in the claims, should not be interpreted as being limited to direct connections only. The terms “coupled” and “connected,” along with their derivatives, may be used. It should be understood that these terms are not intended as synonyms for each other. Thus, the scope of the expression a device A coupled to a device B should not be limited to devices or systems wherein an output of device A is directly connected to an input of device B. It means that there exists a path between an output of A and an input of B which may be a path including other devices or means. “Coupled” may mean that two or more elements are either in direct physical or electrical contact, or that two or more elements are not in direct contact with each other but yet still co-operate or interact with each other.

[0093] Thus, while there has been described what are believed to be the preferred embodiments of the invention, those skilled in the art will recognize that other and further modifications may be made thereto without departing from the spirit of the invention, and it is intended to claim all such changes and modifications as falling within the scope of the invention. For example, any formulas given above are merely representative of procedures that may be used. Functionality may be added or deleted from the block diagrams and operations may be interchanged among functional blocks. Steps may be added or deleted to methods described within the scope of the present invention.

[0094] The above disclosed subject matter is to be considered illustrative, and not restrictive, and the appended claims are intended to cover all such modifications, enhancements, and other

implementations, which fall within the true spirit and scope of the present disclosure. Thus, to the maximum extent allowed by law, the scope of the present disclosure is to be determined by the broadest permissible interpretation of the following claims and their equivalents, and shall not be restricted or limited by the foregoing detailed description. While various implementations of the disclosure have been described, it will be apparent to those of ordinary skill in the art that many more implementations and implementations are possible within the scope of the disclosure. Accordingly, the disclosure is not to be restricted except in light of the attached claims and their equivalents.

Claims

1. A computer-implemented method for collecting and releasing a virtual object across disparate environments using augmented reality, the method comprising: receiving, by an augmented reality application, authentication data including login credentials or biometric data associated with a user; retrieving, by the augmented reality application, a user identifier associated with the user based on the authentication data; generating, by the augmented reality application, a display of a first environment based on a camera input, and detecting an object within the first environment utilizing object detection and classification techniques; transmitting, by the augmented reality application, a search request for the object to a server; receiving, by the augmented reality application, content associated with the object from the server, wherein the content include an option to initiate an electronic transaction; receiving, by the augmented reality application via a transaction interface, an input from the user to initiate the electronic transaction; and generating, by the augmented reality application, a display of a selectable option to collect a virtual object associated with the object.
2. The computer-implemented method of claim 1, further comprising: superimposing, by the augmented reality application, the content over the first environment, wherein the content includes additional information related to the object or one or more sponsor items.
3. The computer-implemented method of claim 1, further comprising: receiving, by the augmented reality application via the transaction interface, transaction information; and transmitting, by the augmented reality application, the transaction information to a merchant system to obtain transaction authorization, wherein the merchant system transmits a transaction authorization request to an acquirer processor server.
4. The computer-implemented method of claim 3, further comprising: receiving, by the augmented reality application, transaction authorization response from the merchant system.
5. The computer-implemented method of claim 1, further comprising: transmitting, by the augmented reality application, a second search request to the server based on the user identifier and the object in response to the selection of the option to collect the virtual object.
6. The computer-implemented method of claim 5, further comprising: receiving, by the augmented reality application, the virtual object or a user library including the virtual object; and storing, by the augmented reality application, the virtual object in a local user library associated with the user identifier.
7. The computer-implemented method of claim 6, further comprising: generating, by the augmented reality application, a display of a second environment based on the camera input; and superimposing, by the augmented reality application, the virtual object over the second environment.
8. The computer-implemented method of claim 6, further comprising: determining, by the server, whether a user library associated with the user identifier is available, wherein the server either retrieves the available user library or generates a new user library.
9. The computer-implemented method of claim 6, wherein the server associates a timestamp or an expiration time with the virtual object based on a transaction amount included in the search request.
10. The computer-implemented method of claim 1, wherein the augmented reality application

confirms an identity of the user by comparing the login credentials or the biometric data with the user identifier.

11. A system for collecting and releasing a virtual object across disparate environments using augmented reality, the system comprising: one or more processors; a data storage comprising instructions which, when executed by the one or more processors, cause the one or more processors to perform operations comprising: receiving, by an augmented reality application, authentication data including login credentials or biometric data associated with a user; retrieving, by the augmented reality application, a user identifier associated with the user based on the authentication data; generating, by the augmented reality application, a display of a first environment based on a camera input, and detecting an object within the first environment utilizing object detection and classification techniques; transmitting, by the augmented reality application, a search request for the object to a server; receiving, by the augmented reality application, content associated with the object from the server, wherein the content include an option to initiate an electronic transaction; receiving, by the augmented reality application via a transaction interface, an input from the user to initiate the electronic transaction; and generating, by the augmented reality application, a display of a selectable option to collect a virtual object associated with the object.

12. The system of claim 11, further comprising: superimposing, by the augmented reality application, the content over the first environment, wherein the content includes additional information related to the object or one or more sponsor items.

13. The system of claim 11, further comprising: receiving, by the augmented reality application via the transaction interface, transaction information; and transmitting, by the augmented reality application, the transaction information to a merchant system to obtain transaction authorization, wherein the merchant system transmits a transaction authorization request to an acquirer processor server.

14. The system of claim 13, further comprising: receiving, by the augmented reality application, transaction authorization response from the merchant system.

15. The system of claim 11, further comprising: transmitting, by the augmented reality application, a second search request to the server based on the user identifier and the object in response to the selection of the option to collect the virtual object.

16. The system of claim 15, further comprising: receiving, by the augmented reality application, the virtual object or a user library including the virtual object; and storing, by the augmented reality application, the virtual object in a local user library associated with the user identifier.

17. The system of claim 16, further comprising: generating, by the augmented reality application, a display of a second environment based on the camera input; and superimposing, by the augmented reality application, the virtual object over the second environment.

18. A non-transitory computer-readable medium storing instructions for collecting and releasing a virtual object across disparate environments using augmented reality, the instructions, when executed by one or more processors, causing the one or more processors to perform operations comprising: receiving, by an augmented reality application, authentication data including login credentials or biometric data associated with a user; retrieving, by the augmented reality application, a user identifier associated with the user based on the authentication data; generating, by the augmented reality application, a display of a first environment based on a camera input, and detecting an object within the first environment utilizing object detection and classification techniques; transmitting, by the augmented reality application, a search request for the object to a server; receiving, by the augmented reality application, content associated with the object from the server, wherein the content include an option to initiate an electronic transaction; receiving, by the augmented reality application via a transaction interface, an input from the user to initiate the electronic transaction; and generating, by the augmented reality application, a display of a selectable option to collect a virtual object associated with the object.

19. The computer-readable medium of claim 18, further comprising: superimposing, by the

augmented reality application, the content over the first environment, wherein the content includes additional information related to the object or one or more sponsor items.

20. The computer-readable medium of claim 18, further comprising: receiving, by the augmented reality application via the transaction interface, transaction information; and transmitting, by the augmented reality application, the transaction information to a merchant system to obtain transaction authorization, wherein the merchant system transmits a transaction authorization request to an acquirer processor server.
